

Leveraging AR and Object Interactions for Emotional Support Interfaces

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ABSTRACT

This work explores ways we can leverage augmented reality systems to create meaningful and emotional interactions with physical objects. We explore how hand and eye interactions can be used to trigger meaningful sensory feedback to the user. These investigations show how AR interfaces can interpret the actions of a user to shift the primary purpose of a physical object for emotional support.

Index Terms: Human-centered computing—Tangible Interfaces—Augmented Reality—Object Interaction; —Emotional Interfaces—

1 INTRODUCTION

Emotional health is integral to one's overall well-being. Good mental health allows people to be productive and tackle the challenges they may encounter in their daily lives. Furthermore, studies show that there are many physical benefits to improved emotional health including lower blood pressure, healthier weight, and a reduced risk of heart disease [1]. An effective way for one to maintain their mental health is to connect with friends and family. Our loved ones are often an indication of the positive support we have in our lives. Objects often help us to find emotional support when loved ones are not around. Sentimental objects serve as mementos of cherished memories with them. We hold on to these objects not because they have a functional purpose but for their sentimental value and emotional output. They serve as souvenirs of our treasured moments with old friends, family, romantic partners, or places we have visited. Especially in the isolating times of the ongoing pandemic, these tokens of our bonds are essential for upholding our mental health.

Our perception of emotional objects are heightened by their affordances, or what their tangible and intangible properties have to offer us [2]. Our visual perception of these mementos is enough to spark nostalgic feelings, but they can also be heightened by touch and interaction. The way we may hold an object or the sensations we get from feeling its unique textures can also invoke emotional reactions. For example, gripping a baseball in a specific way or feeling the softness of a pillow. In MusicBottles, Ishii et al. leverages both the visual and tangible properties of glass bottles to output different musical sounds and provide an emotional aspect to the interaction. This allows the simple action of opening a glass bottle to become a way of collecting memories or feelings of nostalgia from the object [4]. Fundamentally, adding new virtual meanings to these inert objects allow us to extend their affordances into the digital world.

Most Augmented Reality (AR) applications are focused on pragmatic tasks- for instance, to display information to the user in an unobtrusive manner, or to simulate how a room would look like after a renovation. With the help of environment reconstruction, objects can also react realistically with the environment. For example, a virtual ball can fall off of a physical table in the user's environment. In other cases, applications use virtual augmentations on a physical object in order to make it appear more interesting. However, in all of

these cases, the fundamental idea revolves around interacting strictly with virtual objects to alter our visual perception of the world.

As AR becomes more and more ubiquitous, it provides an opportunity to explore new ways to support our well-being. In our work, we invert the focus from the virtual to the physical and use interactions with physical objects to trigger virtual and auditory outputs to augment our perception of the object and the environment. Now, inanimate objects can be brought to life and given a new meaning. In this paper we discuss how AR can be used in emotional design and describe the implementation of the Nostalgic Trophy.

2 DESIGNING FOR EMOTIONAL INTERACTION

The goal of emotional interaction is to go beyond the functionality and connect with the user at a deeper level. One approach is to understand the interface in terms of sensory signals that are processed at the visceral, behavioral, and reflective cognitive levels. To evoke more of an emotional response from the user, the designer start by analyzing the context and needs of the user [3].

In this work, we focus on how an emotional interface can reestablish connections with a user's past memories. Emotional design is applied by capturing the intention and needs of the user - in this case, the need to reconnect with their past- and can tailor their immersive experience to help them achieve the comfort they seek. The AR system can act with such advanced personalizing as it essentially sees from the user's point of view.

Various modalities can be used to detect and enrich the interactions between users and objects: gaze can be used for transparent selection, gestures can be used to identify uses, and touch can be used to explore tangible aspects. Some example scenarios include: throwing an object down would capture frustration, which would trigger calming music and graphics to alleviate their stress; pulling the frame of a picture of your family closer to you could prompt an offer to call the person in the photo.

2.1 Nostalgic Trophy Interaction

Trophies are objects with strong emotional content associated with it. For the nostalgic trophy our goal was to explore how to allow users to experience pieces of wistful immersion based on the intensity of the interaction. Subtle interactions should provide non-intrusive feedback while active interactions could be more intense and vivid. For this exploration, we chose to augment interactions with an old high school cheerleading trophy. The Nostalgic Trophy explores three separate object interactions:

1. **Gaze:** When gazing at the trophy, the user hears a football stadium audio of crowds chanting and cheering. This feedback mildly immerses the user in a game-day setting to provoke a feeling of nostalgia.
2. **Touch:** When touching the trophy with any finger (or multiple), the user sees a sparkling animation appear around the trophy, reminding the user of fireworks and other celebratory elements that came with cheerleading (Figure 3).
3. **Movement:** When the user moves the trophy either closer or away from them, a graphic of a football field and football appears around the trophy. The goal is to increase the sense of connectiveness to the event (Figure 2).

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We created a proof-of-concept of this general idea using Microsoft HoloLens2 and the Mixed Reality Toolkit (MRTK) v2, along with Vuforia's Object Target recognition. Gaze interaction was implemented by checking whether the trophy was tracked in the HoloLens' field of vision. To detect where the user touched the physical object, we used MRTK's hand tracking and environment reconstruction to perform a raycast from each fingertip to the object mesh. If the finger tip was within the specified threshold, then that finger was considered to be in contact with the object (Figure 1). Finally, movement of the user relative to the trophy was determined by checking if the trophy's world coordinates changed by a certain threshold from where it was last seen.

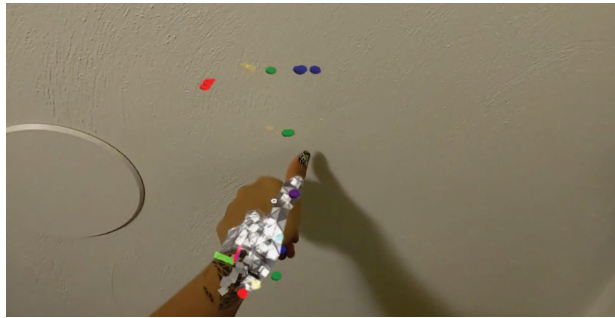


Figure 1: Testing the accuracy of finger-object detection. The user touches the wall with a finger and a marker is left on the surface.

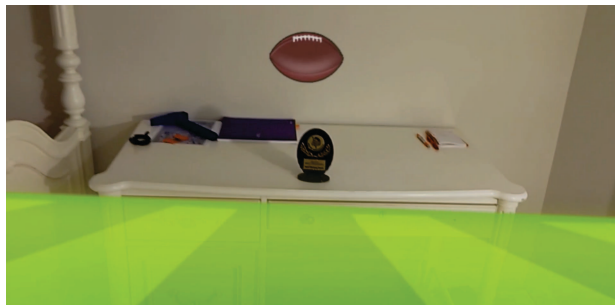


Figure 2: Trophy Movement Interaction. When the user moves the trophy, football-related graphics appear around the trophy

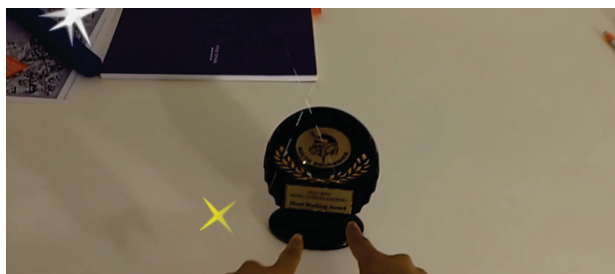


Figure 3: Trophy Touch Interaction. Sparkling animation appear around the trophy, reminding the user of fireworks and other celebratory elements

2.2 Discussion and Future Work

Ultimately, the AR app is able to sense different intentions while interacting with the physical object and provide an appropriate feed-

back in return. The interactions allowed for increased feedback depending on how engaged the user was with the object. If it is a simple gaze at the object target, the feedback is audio only, but if touch or movement is detected in the interaction a corresponding visual component is displayed in response.

During the preliminary interface evaluations, we found that users found the increasing intensity of sensory feedback to be appropriate with how they interacted with the trophy. We were able to find a positive correlation between showing happy memories and eliciting happy emotions from the user. However, the overall feel of the application could be made smoother by making the touch interactions more realistic. We found it difficult to confirm the intention of the user with this implementation. When touching an object with the index finger using a pointing gesture, the inactive fingers would sometimes also be tracked as "touching" the object because they were close to the object mesh. Since it was the user's intention to only make contact with their index finger, we intend to improve track the gesture of the user's hand and then determine which fingers are "enabled" for interacting. Furthermore, the application lacked a natural feel when the animation would render before the user touched the trophy. This could be improved by using the velocity the hand to extrapolate the hand/finger movement a few milliseconds into the future once the hand stops moving. This way the graphics would render at the moment the user's finger moves away from the trophy.

The current implementation does not leverage eye tracking since we only needed to identify if the object was in the field of view. However, this could be used to detect gaze at specific parts of an object. Positional audio could also be used to decrease in volume if you walk away from the trophy. This would improve the perception that the sounds and memories are emanating from the meaningful object. Lastly, this application can be paired with external services in order to offer other actions such as calling a person related to the object or posting a note about these memories on a social media platform. Access to a user's photo libraries could allow us to replay their actual memories before them to increase the interface's effectiveness.

3 CONCLUSION

Our goal was to explore ways that interacting with physical objects can extend user's emotional perception of objects. By leveraging mixed reality and object recognition, we were able to display scenarios where users can explore a new dimension as a result of a physical interaction. Furthermore, we were able to portray these responses in a way that would spark feelings of nostalgia from the user and therefore tie in emotional support to the application's use case. In the future, such an application can be used to help users reflect on their past and appreciate the passage of time as life goes on. Specifically, it could support individuals who may live alone or away from their family. For example, an elderly person or a college student, and help them still feel connected with their past by enhancing the emotional feelings that are tied to their sentimental belongings.

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